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Q1.

$2x^2 = 1 + x^2$

$x^2 = 1 \quad , x = -1 \quad ; x = 1$

$$\int_{-1}^{1}\int_{2x^2}^{1+x^2} x + 2y \; dy\, dx$$

$$\int_{-1}^{1} xy + y^2 \Big|_{2x^2}^{1+x^2} dx$$

$$\int_{-1}^{1} x(1+x^2-2x^2) + (1+x^2)^2 - 4x^4 \; dx$$

$$= \int_{-1}^{1} -3x^4 - x^3 + 2x^2 + x + 1$$

$$-\frac{3}{5}x^5 - \frac{1}{4}x^4 + \frac{2}{3}x^3 + \frac{1}{2}x^2 + x \Big]_{-1}^{1}$$

$$\left(-\frac{3}{5} - \frac{1}{4} + \frac{2}{3} + \frac{1}{2} + 1\right) - \left(\frac{3}{5} - \frac{1}{4} - \frac{2}{3} + \frac{1}{2} - 1\right)$$

$$= \frac{32}{15}$$

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b) $x^2 = 2x \Rightarrow x(x-2)=0 \Rightarrow x=0,\ x=2$

$$V = \int_0^2 \int_{x^2}^{2x} (x^2+y^2)\,dy\,dx$$

$$= \int_0^2 x^2 y + \frac{1}{3}y^3 \Big|_{x^2}^{2x} dx$$

$$= \int_0^2 x^2(2x-2x^2) + \frac{1}{3}(8x^3 - x^6)\,dx$$

$$\int_0^2 \frac{14}{3}x^3 - x^4 - \frac{1}{3}x^6\,dx$$

$$\frac{14}{12}x^4 - \frac{1}{5}x^5 - \frac{1}{21}x^7 \Big]_0^2$$

$$\frac{7}{6}(16) - \frac{32}{5} - \frac{128}{21} = \frac{216}{35}$$

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Q2.

a) $\int_0^3 \int_{-1}^2 \int_0^1 x y z^2 \, dx \, dy \, dz$

$\int_0^1 x \, dx \int_{-1}^2 y \, dy \int_0^3 z^2 \, dz$

$\left.\frac{x^2}{2}\right|_0^1 + \left.\frac{y^2}{2}\right|_{-1}^2 + \left.\frac{z^3}{3}\right]_0^3$

$= \frac{1}{2} \times \frac{3}{2} \times 9 = \frac{27}{4}$

b) $\int_0^1 \int_0^{y^2} \int_1^{z+1} x^{-3} \, dx \, dz \, dy$

$\int_1^{z+1} x^{-3} = \left.\frac{x^{-2}}{-2}\right|_1^{z+1} = -\frac{1}{2}\left(\frac{1}{(z+1)^2} - 1\right)$

$\int_0^{y^2} \frac{1}{2} - \frac{1}{2(z+1)^2} \, dz = \left[\frac{1}{2} z + \frac{1}{2(z+1)}\right]_0^{y^2}$

$= \frac{y^2}{2} + \frac{1}{2(y^2+1)} - \frac{1}{2}$

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$$\int_0^1 \frac{y^2}{2} + \frac{1}{2(y^2+1)} - \frac{1}{2}\,dy$$

$$\frac{y^3}{6} + \frac{1}{2}\tan^{-1}(y) - \frac{y}{2}\Big|_0^1$$

$$= \frac{1}{6} + \frac{1}{2}\cdot\frac{\pi}{4} - \frac{1}{2} - 0 = \frac{\pi}{8} - \frac{1}{3}$$

Q3.

a) $x(t) = 2 + (3-2)t = 2+t$

$dx = dt$

$y(t) = 0 + (4-0)t = 4t$

$dy = 4\,dt$

$z(t) = 0 + (5-0)t = 5t$

$dz = 5dt$

$t \in [0, 01]$

$$\int_0^1 (4t)(1) + (5t)(4) + (2+t)5\,dt$$

$$= \int_0^1 4t + 20t + 10 + 5t\,dt$$

$$\int_0^1 29t + 10\,dt = \frac{29t^2}{2} + 10t\Big|_0^1$$

$$= \frac{29}{2} + 10 = 24.5$$

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C_2:

$x = 3$ $\quad dx = 0\,dt$

$y = 4$ $\quad dy = 0\,dt$

$dz = dz$

$$\int_5^0 (4)(0) + (2)(0) + 3\,dz = \int_5^0 3\,dz$$

$$3z\Big|_5^0 = 0 - 15 = -15$$

$$\int_c = 24.5 - 15 = 9.5$$

b) $x = t$ $\quad dx = dt$

$y = t^2$ $\quad dy = 2t\,dt$

$ds = \sqrt{1^2 + 4t^2}\,dt$ $\quad t \in [0,1]$

$$\int_1^0 2t\sqrt{1+4t^2}\,dt$$

$$\frac{1}{4}\int_1^0 8t\sqrt{1+4t^2}\,dt = \frac{1}{4}\left[\frac{2}{3}(1+4t)^{3/2}\right]_0^1$$

$$= \frac{1}{4}\left[\frac{2}{3}(5)^{3/2} - \frac{2}{3}(1)^{3/2}\right]$$

$$= \frac{1}{6}(5\sqrt{5} - 1)$$

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C_2:

$x(t) = 1$ $\qquad dx = 0dt$

$y(t) = 1 + (2-1)t = 1+t$

$dy = dt$

$ds = \sqrt{0^2 + 1^2} = 1$

$ds = dy$ (since $dx = 0$)

$$\int_1^2 2(1)\, dy = 2y\Big]_1^2 = 4 - 2 = 2$$

$$\int_c = \frac{1}{6}(5\sqrt{5} - 1) + 2 = \frac{5\sqrt{5} + 11}{6}$$

Q4:

$P = x^4 = \frac{\partial P}{\partial y} = 0$

$Q = xy = \frac{\partial Q}{\partial x} = y$

$$\iint_D (y - 0)\, dA = \int_0^1 \int_0^{1-x} y\, dy\, dx$$

$$\int_0^{1-x} y\, dy = \frac{y^2}{2}\Big|_0^{1-x} = \frac{(1-x)^2}{2} - 0$$

$$\frac{1}{2}\int_0^1 1 - 2x + x^2\, dx = \frac{1}{2}\left[x - x^2 + \frac{1}{3}x^3\right]_0^1$$

$$= \frac{1}{2}\left(1 - 1 + \frac{1}{3}\right) = \frac{1}{6}$$

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b) $P = 3y - e^{\sin x} \Rightarrow \frac{\partial P}{\partial y} = 3$

$Q = 7x + \sqrt{y^4+1} \Rightarrow \frac{\partial Q}{\partial x} = 7$

$\iint_D 7-3 \, dA = \iint_D 4 \, dA$

$D = \pi R^2 = 9\pi$

$9\pi \times 4 = 36\pi$

Q5 a)

$i: \frac{\partial}{\partial y}(-y^2) - \frac{\partial}{\partial z}(xy) = -2y - 0 = -2y$

$j: -\left[\frac{\partial}{\partial x}(-y)^2 - \frac{\partial}{\partial z}(xz)\right] = -[0-x] = x$

$k: \frac{\partial}{\partial x}(xy) - \frac{\partial}{\partial y}(xz) = y - 0 = y$

$\text{curl } F = -2yi + xj + yk$

$\text{div } F = \frac{\partial}{\partial x}(xz) + \frac{\partial}{\partial y}(xy) + \frac{\partial}{\partial z}(-y^2)$

$\text{div } F = z + x + 0 = z + x$

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b) $P = xy$ $\quad \frac{\partial P}{\partial y} = x$

$Q = x^2$ $\quad \frac{\partial Q}{\partial x} = 2x$

$$\int_D 2x - x \, dA \Rightarrow \int_D x \, dA$$

$$\int_0^1 \int_0^{1-x} x \, dy \, dx \qquad \begin{array}{l} 0 \le x \le 1 \\ 0 \le y \le 1-x \end{array}$$

$$\int_0^1 xy \Big|_0^{1-x} dx = \int_0^1 x(1-x) \, dx = \int_0^1 x - x^2 \, dx$$

$$\frac{x^2}{2} - \frac{x^3}{3} \Big|_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$